

EBNAT-KAPPEL, Switzerland

WMO: 066930

Lat: 47.2733N

Lon: 9.1086E

Elev: 623

StdP: 94.06

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.1	-10.1	-15.4	1.1	-11.2	-12.1	1.4	-8.3	5.2	4.0	4.4	1.3	1.3	150	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	29.6	20.2	27.7	19.3	25.9	18.7	21.1	28.1	20.1	26.4	19.2	25.0	2.1	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.6	24.3	17.7	13.7	22.9	17.0	13.1	21.8	64.0	28.2	60.4	26.5	57.2	25.3	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.6	4.0	3.4	DB	-17.2	32.1	3.5	1.1	-19.7	32.9	-21.7	33.6	-23.7	34.2	-26.2	35.1	(4)
(5)			WB	-17.4	22.8	3.4	1.0	-19.9	23.5	-21.9	24.1	-23.8	24.6	-26.3	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-0.3	-0.5	4.4	8.4	11.8	16.2	17.8	17.7	13.4	9.4	4.3	1.0
	DBStd	7.45	4.21	4.94	3.35	3.60	3.42	3.31	3.08	3.03	2.87	3.57	3.79	3.55
	HDD10.0	1389	319	293	176	71	21	1	0	0	4	54	173	279
	HDD18.3	3598	577	527	433	297	205	80	47	50	148	277	421	537
	CDD10.0	911	0	0	1	24	75	187	241	238	107	35	3	0
	CDD18.3	77	0	0	0	0	1	17	29	29	1	0	0	0
	CDH23.3	1084	0	0	0	6	37	271	388	349	32	1	0	0
	CDH26.7	279	0	0	0	0	3	75	111	91	0	0	0	0
Wind	WSAvg	1.1	1.1	1.1	1.2	1.3	1.2	1.2	1.1	1.0	0.9	0.9	0.9	0.9
Precipitation	PrecAvg	1530	92	96	109	109	161	172	173	166	127	104	110	113
	PrecMax	1972	209	272	257	210	351	261	322	285	239	207	240	268
	PrecMin	1206	7	29	38	18	43	71	65	29	58	8	0	5
	PrecStd	176	56	54	51	50	70	51	66	65	46	47	56	59
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	13.4	18.3	23.9	26.8	31.5	32.3	31.1	25.7	21.8	18.0	13.6
		MCWB	8.3	7.0	10.2	14.0	16.9	21.2	21.0	20.3	18.7	13.7	14.5	6.4
	2%	DB	8.0	10.1	16.0	21.4	23.9	28.8	29.5	29.1	24.3	19.6	15.3	9.8
		MCWB	5.9	5.9	9.2	12.7	15.7	19.9	20.0	20.3	17.8	14.0	10.8	6.7
	5%	DB	6.2	7.6	13.9	19.7	21.7	26.6	27.4	27.2	22.1	17.6	12.4	7.9
		MCWB	4.8	4.5	8.4	12.2	14.3	18.9	19.2	19.7	16.8	13.3	9.7	6.0
	10%	DB	4.7	5.4	11.4	17.2	19.9	24.1	25.2	25.0	19.9	15.2	9.9	6.1
		MCWB	3.9	3.4	7.2	11.0	13.4	17.6	18.2	19.2	15.8	12.5	8.3	4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	9.0	11.1	14.7	18.1	22.6	22.0	21.8	19.4	16.0	13.3	8.7
		MCDB	11.0	11.9	16.1	21.6	25.0	29.9	29.6	28.2	24.7	19.5	17.7	11.4
	2%	WB	6.2	6.5	9.8	13.5	16.5	20.6	20.8	20.9	18.2	14.6	11.0	7.4
		MCDB	7.6	9.2	14.3	19.9	22.5	27.1	28.4	27.6	22.9	17.9	13.9	9.3
	5%	WB	5.1	5.0	8.7	12.5	15.2	19.2	19.8	20.1	17.3	13.7	9.7	6.0
		MCDB	6.0	7.5	12.9	18.6	20.3	25.7	26.0	26.3	21.6	16.6	12.2	8.0
	10%	WB	4.0	3.6	7.6	11.4	14.1	18.1	18.7	19.2	16.1	12.7	8.5	4.9
		MCDB	4.7	5.1	11.3	16.8	18.9	23.4	24.3	24.8	19.5	15.0	10.1	6.3
Mean Daily Temperature Range	5% DB	MDBR	6.7	8.0	10.7	11.9	11.2	11.3	11.4	11.6	10.7	9.7	7.8	7.6
		MCDBR	8.0	11.9	16.2	17.7	16.4	16.6	15.8	15.8	14.4	14.1	12.4	10.1
		MCWBR	6.1	7.3	9.8	9.7	8.8	8.2	7.2	7.8	8.5	8.6	8.2	7.1
	5% WB	MCDBR	7.5	10.6	14.7	16.6	14.4	15.2	14.5	14.7	13.4	12.4	11.4	9.4
MCWBR		5.9	7.0	9.2	9.7	8.0	7.9	7.2	7.6	8.2	8.1	7.8	6.9	
Clear-Sky Solar Irradiance	taub	0.278	0.304	0.342	0.381	0.396	0.403	0.405	0.397	0.367	0.338	0.306	0.275	
	taud	2.401	2.339	2.316	2.278	2.286	2.308	2.299	2.339	2.402	2.468	2.487	2.429	
	Ebn at Noon	816	861	876	871	868	862	854	846	841	814	777	776	
	Edh at Noon	68	90	108	124	127	126	125	115	99	79	64	59	
All-Sky Solar Radiation	RadAvg	1.24	2.11	3.32	4.36	4.95	5.70	5.55	4.84	3.65	2.32	1.32	1.02	
	RadStd	0.14	0.28	0.45	0.64	0.64	0.50	0.53	0.50	0.48	0.37	0.26	0.17	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (122 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.70	+0.42	+0.33	N/A	N/A	+57	+15

Nomenclature: See separate page